

Advanced (Habanero)

- Q1.** What is the absolute value of x if $x = -5$?
(A) -5
(B) 5
(C) 0
(D) 10
(E) -10
- Q2.** If $|x| = 7$, then what are the possible values of x ?
(A) 7 only
(B) -7 only
(C) 7 or -7
(D) 0 or 7
(E) 0 or -7
- Q3.** What is $|-3| + |2|$ equal to?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9
- Q4.** If $|x + 2| = 4$, what are the possible values of x ?
(A) 2 or -6
(B) -2 or 6
(C) 2 or 6
(D) -6 only
(E) 6 only
- Q5.** What is $|0|$ equal to?
(A) 1
(B) 0
(C) -1
(D) -0
(E) Not defined
- Q6.** What does $|x - 3|$ represent?
(A) The distance between x and 3 on the number line
(B) The sum of x and 3
(C) The difference between x and 3
(D) The product of x and 3
(E) The quotient of x and 3
- Q7.** If $|x| > 2$, what can be concluded about x ?
(A) x is between -2 and 2
(B) x is less than -2 or greater than 2
(C) x is greater than 2 only
(D) x is less than -2 only
(E) x is equal to 0
- Q8.** What is the value of $|x|$ if $x = 9$?
(A) -9
(B) 9
(C) 0
(D) 1
(E) -1

Open Ended Questions (FRQ)

- Q9.** A company packs boxes with x items in each box. If the absolute value of the difference between the number of items packed and 15 is less than 3 , how many items can be packed in each box?
- Q10.** If $|2x - 5| = 3$, find the values of x and explain the meaning of the absolute value in this context.

Chef's Tips

- Tip 1: Ensure students understand the definition of absolute value as distance from zero on the number line.
- Tip 2: Emphasize the importance of considering both positive and negative solutions when dealing with absolute value equations.
- Tip 3: Encourage students to use real-world examples to illustrate their understanding of absolute value concepts.